

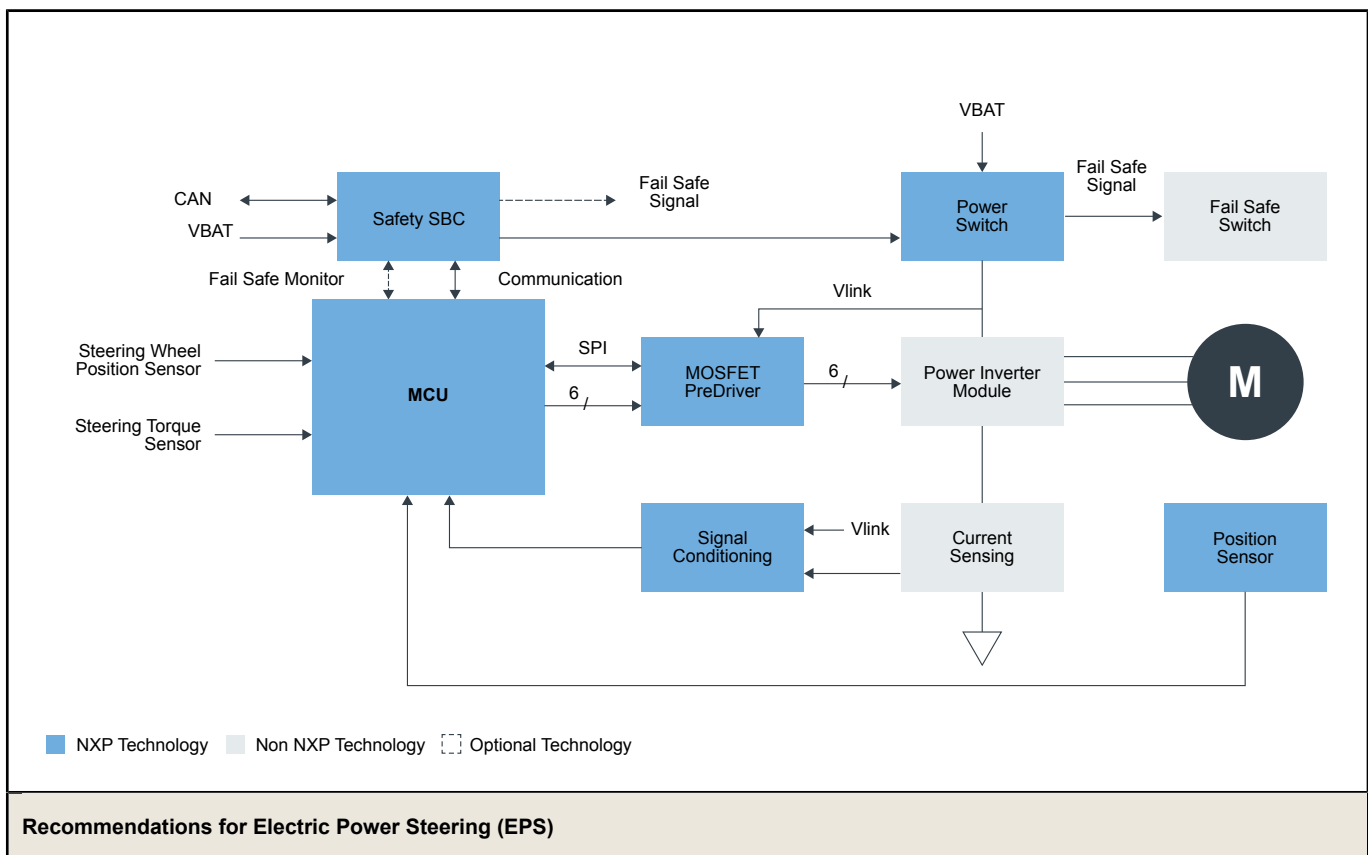


Electric Power Steering (EPS)

Last Updated: Jul 15, 2026

Electric Power Steering (EPS) offers improved handling and steering feel while improving vehicle safety by adopting the steering torque to the vehicle speed and providing active torque in critical driving situations. Our 16-bit and 32-bit single and dual-core automotive MCUs provide enhanced computing power and specialized peripherals for complex electric motor control functions. Integrated power supply solutions are also important elements of a power steering control unit. They provide connectivity to automotive busses, such as CAN and LIN. For MOSFET power stages control, integrated pre-drivers are typically used to interface with the MCU directly or via SPI. For MOSFET power stages control, integrated pre-drivers are typically used to interface with the MCU directly or via SPI.

Electric Power Steering (EPS) Block Diagram



Microcontrollers (MCU)	• S32K Automotive General-Purpose Microcontrollers • S32K39-37-36: S32K39/37/36 Microcontrollers for Electrification Applications • S32Z and S32E Real-Time Processors • S32E2: S32E2 Safe and Secure High-Performance Real-Time Processors with Actuation Support
Safety SBC	• High Speed CAN with Flexible Data Rate (CAN FD) • TJA144x: Automotive CAN FD Transceiver Family • 5 V PMIC Solutions • PF5030: Multi-Channel PMIC for Automotive Applications • 12 V, 24 V, and 48 V SBC Solutions • VR5510: Multi-Channel (9) PMIC for S32G Processor – 8 High Power, 1 Low Power, Fit for ASIL D Safety Level • FS86: Safety System Basis Chip For Domain Controller, Fit For ASIL B and D • FS26: Safety System Basis Chip with Low Power, for ASIL D Systems • Automotive Ethernet PHY Transceivers • TJA1103: TJA1103, ASIL B Compliant Automotive Ethernet 100BASE-T1 PHY Transceiver • CAN Signal Improvement Capability (SIC) • TJA1462: CAN Signal Improvement Capability Transceiver with Standby Mode • Secure TJA115x CAN Transceiver Family • TJA1152: Secure HS-CAN Transceiver with Standby Mode • S32Z and S32E Real-Time Processors • S32E2: S32E2 Safe and Secure High-Performance Real-Time Processors with Actuation Support
MOSFET Pre-Driver	• BLDC, H-Bridge and Stepper Motor Drivers • MC33937: 3-Phase Field Effect Transistor Pre-Driver • GD3000: 3-Phase Brushless Motor Pre-Driver
Signal Conditioning	• Multi-Switch Detection Interfaces (MSDI) • MC33972: MSDI with Suppressed Wakeup • CD1020: Low-Cost 22-CH Multiple Switch Detect Interface

View our complete solution for [Electric Power Steering \(EPS\)](#).

Note: The information on this document is subject to change without notice.

www.nxp.com

NXP and the NXP logo are trademarks of NXP B.V. All other product or service names are the property of their respective owners. The related technology may be protected by any or all of patents, copyrights, designs and trade secrets. All rights reserved. © 2026 NXP B.V.